MILITARY SPECIFICATION

METHENAMINE HIPPURATE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Methenamine Hippurate Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.4).

STANDARDS

Federal

FED-STD-140	Tablets (for Medicinal Purposes).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Methenamine Hippurate Tablets, and shall be in accordance with all applicable requirements of FED-STD-140 as specified herein:

S2. Classification. Shall be type I, class 1.

S5.3 The following additional requirements and tests are added to this paragraph:

The tablets shall contain in each tablet 1.0 Gram of methenamine hippurate, within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of methenamine hippurate, when assayed as specified in 4.3.1.1.

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The tablets shall comply with the identity tests specified in 4.3.1.2.

S6.4.6 Hardness. Tablets shall have a hardness of not less than 7 kg and not more than 13 kg when tested using the Pfizer Hardness Tester.

S6.4.9.1 Disintegration. Shall meet the requirements of the U.S.P. Disintegration Test for Uncoated Tablets in not more than 30 minutes, using U.S.P. simulated gastric fluid T.S.

S6.4.10.1 Uncoated tablets. The tablets shall comply with the requirements of the U.S.P. Weight Variation Test.

3.2 Ingredients.

3.2.1 Methenamine Hippurate. The Methenamine Hippurate used in the manufacture of the tablets shall comply with the following requirements:

3.2.1.1 Description. Shall be a white, crystalline powder having an immediate sour taste, a mildly bitter aftertaste, and practically no odor.

3.2.1.2 Clarity and color of solution. A 10 percent solution in 95 percent ethanol shall be clear and its absorbance at 436 mu in a 1-cm path length shall not be greater than 0.030.

3.2.1.3 Loss on drying. The powder shall lose not more than 1 percent of its weight when dried for one (1) hour in vacuum (2 mm Hg), at a temperature of 60°C., using a suitable apparatus.

Proceed with the following tests on material dried according to the above test for loss on drying.

3.2.1.4 Identification. The infrared absorption spectrum (graph) from 2.5 to 15u of a Nujol mull containing a 20 percent sample concentration, shall compare qualitatively to an infrared absorption spectrum (graph) of a reference standard similarly prepared, as kept on file by the manufacturer. A copy of the standard infrared spectrum (graph) shall be submitted by the contractor, and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

3.2.1.5 Assay for methenamine. Shall contain not less than 42.0 percent and not more than 46.0 percent methenamine when determined in 4.3.2.1.

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3.2.1.6 Assay for hippuric acid. Shall contain not less than 54.0 percent and not more than 58.0 percent hippuric acid when determined in 4.3.2.2.

3.2.2 The methenamine used in the preparation of the methenamine hippurate shall be in accordance with the tests, standards, and requirements of the N.F., including any supplements or revisions thereto.

3.2.3 The hippuric acid used in the preparation of the methenamine hippurate shall comply with the following:

3.2.3.1 Description. Shall be a white to off-white, odorless, crystalline powder.

3.2.3.2 Benzoic acid. Shall contain not more than 1.0 percent benzoic acid, when determined in 4.3.2.3.

3.2.3.3 Residue on ignition. Shall be not more than 1.0 percent determined by the U.S.P. method.

3.2.3.4 Heavy metals. Shall not exceed 20 parts per million (p.p.m.) when determined in 4.3.2.4.

3.2.3.5 Loss on drying. The powder shall lose not more than 1 percent of its weight when dried over P_2O_5 at room temperature in vacuum (1 mm Hg) for 18 hours.

Proceed with the following tests on material dried according to the above test for loss on drying.

3.2.3.6 Melting range. The powder shall melt between 187° C. and 192° C. with decomposition when determined by the U.S.P. Class 1a method.

3.2.3.7 Identification.

(a). I.R. spectrum. The infrared absorption spectrum (graph) from 2.5 to 15 u of a Nujol mull containing a 20 percent sample concentration shall compare qualitatively to an infrared absorption spectrum (graph) of a reference standard similarly prepared as kept on file by the manufacturer.

(b). U.V. spectrum. The ultraviolet absorption spectrum (graph) of the sample, from 200 to 400 mu, at a concentration of 0.0100 mg/ml in ethyl alcohol, shall compare qualitatively to an ultraviolet absorption spectrum (graph) of a reference standard similarly prepared as kept on file by the manufacturer.

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A copy of the standard infrared spectrum (graph) and of the standard ultraviolet spectrum (graph) shall be submitted by the contractor, and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

3.2.3.8 Assay for hippuric acid. Shall contain not less than 98.0 percent and not more than 102.0 percent hippuric acid when determined in 4.3.2.2 except the accurately weighed sample shall be 110 mg of hippuric acid.

3.3 Potency Period. The potency period (Expiration dating period) shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records or examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 5 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be in accordance with Federal Standard Fed. Std. No. 140a.

4.3 Tests.

4.3.1 Tests for finished tablets.

4.3.1.1 Assay for methenamine hippurate in the tablets.

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and one ml methanol. Spot 10 ml of this solution on the plate

Accurately weigh 20 tablets, selected at random, to the nearest milligram and place then in the jar of a suitable blender. Add approximately 250 ml of 95 percent ethanol to the jar and cover it with a suitable cap. Blend at a low speed until the tablets have been partially disintegrated. Rinse down the sides and cap of the jar with 95 percent ethanol and blend again at a medium speed for 5 minutes. Quantitatively transfer the blend to a 500-ml volumetric flask, dilute to mark with 95 percent ethanol and mix. Pipet 4.00 ml of the solution into each of two 50-ml Erlenmeyer flasks (these are the sample aliquots). Add 4 ml of 95 percent ethanol and 5 ml of purified water to a 50-ml Erlenmeyer flask (this is the blank).

Treat the samples and the blank identically as follows:

(a). Add 10 drops of thymolphthalein indicator (0.5 percent w/v thymolphthalein in 95 percent ethanol).

(b). Titrate with standard, 0.1N sodium hydroxide to a blue or greenish-blue end point, and record the volume (ml) used. Note: The end point of the samples is the first color change to greenish-blue; the end point of the blank is the first faint bluish color.

Calculate the methenamine hippurate content for each of the two samples by the following formula:

Methenamine hippurate content (mg per tablet) =

$$\frac{(Vs-Vb)(Nb)\ (319.4)(500)}{(20)\ (4)} =$$

$$(Vs-Vb)\ (Nb)\ (1996)$$

Where,

Vs and Vb = the volumes of 0.1N sodium hydroxide used for the sample aliquot and blank, respectively (ml)

Nb = The exact normality of the standard, 0.1N sodium hydroxide (meq/ml)

319.4 = the milliequivalent weight of methenamine hippurate (mg/meq)

500 = the total sample volume (ml)

4 = the sample aliquot volume (ml)

20 = the number of tablets taken for the sample (tablet).

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4.3.1.2 Identification of the finished tablets. Triturate approximately 750 mg of powdered tablet with 15 ml of purified water. Filter the triturated tablet through No. 3 Whatman paper and collect the filtrate in a 50-ml Erlenmeyer flask. Add 10 ml of 10 percent sulfuric acid to the flask. Moisten a piece of filter paper with U.S.P. silver-ammonium nitrate T.S. and while holding the paper in the neck of the flask, heat the solution. When the paper turns brown to black and the characteristic odor of formaldehyde is detected, proceed as follows:

a. Add 2 ml of 50 percent sodium hydroxide to the warm mixture while holding a piece of moistened red litmus paper in the neck of the flask.

b. The litmus paper shall turn blue, indicating the presence of ammonia.

4.3.2 Tests for ingredients.

4.3.2.1 Assay for methenamine.

Reagents:

Glacial acetic acid, reagent grade.
Potassium acid phthalate, primary standard.
Perchloric acid, standard, 0.025N in glacial acetic acid - Dilute 2.1 ml of 70% perchloric acid and 5.0 ml of acetic anhydride to 1000 ml with glacial acetic acid. Standardize by the assay procedure, using about 20 mg of potassium acid phthalate.
Crystal violet indicator - Dissolve 20 mg of crystal violet in 100 ml of glacial acetic acid.

Procedure:

Dissolve about 30 mg of methenamine hippurate, accurately weighed, in 2.0 ml of glacial acetic acid. Add 2 drops of crystal violet indicator and titrate with standard, 0.025N perchloric acid in glacial acetic acid to the disappearance of the violet color. Titrate a 2.0-ml glacial acetic acid blank in the same manner.

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Calculations:

$$\frac{(140.2)(N)(Vs-Vb)(100)}{(W)} = \text{percent methenamine content}$$

where,

140.2 = equivalent weight of methenamine (mg/meq)
N = exact normality of perchloric acid (meq/ml)
Vs = volume of perchloric acid used for sample (ml)
Vb = volume of perchloric acid used for blank (ml)
100 = percentile factor
W = sample weight (mg)

4.3.2.2 Assay for hippuric acid.

Reagents and apparatus:

95% Ethanol (Alcohol, U.S.P.)
Sodium hydroxide, standard 0.1N
Recording potentiometric titrator equipped with a glass-silver-silver chloride or glass-calomel electrode pair.

Procedure:

Weigh accurately about 200 mg of methenamine hippurate into a 50-ml beaker. Add 30 ml of 95 percent ethanol and stir magnetically to dissolve the sample. Adjust the recorder to display the pH range 3 to 13 as about full scale. Perform the titration with standard, 0.1N sodium hydroxide and record the volume required to reach the end point, which is the center of the major inflection. Titrate a blank consisting of 30 ml of 95 percent ethanol, plus purified water approximately equal in volume to the titrant required to reach the end point in the sample titration. Record the volume of titrant required to reach the pH of the sample end point.

Calculations:

$$\frac{(179.2)(N)(Vs-Vb)(100)}{(W)} = \text{percent hippuric acid content}$$

where,

179.2 = equivalent weight of hippuric acid (mg/meq)
N = exact normality of sodium hydroxide (meq/ml)
Vs = volume of sodium hydroxide used for the sample (ml)
Vb = volume of sodium hydroxide used for the blank (ml)
100 = percentile factor
W = sample weight (mg).

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4.3.2.3 Assay for benzoic acid in hippuric acid.

Weigh accurately about 1 gram of the hippuric acid powder into a 35-ml, glass-stoppered centrifuge tube. Add 5.00 ml of 5N hydrochloric acid and 10.00 ml of benzene, and shake mechanically for 5 minutes. Transfer 5.00 ml of the benzene phase into a 60-ml separatory funnel, add 20.00 ml of 0.1N sodium hydroxide, and shake vigorously for 30 seconds. Allow the phases to separate completely and transfer 5.00 ml of the aqueous phase to a 10-ml volumetric flask. Heat on a steam bath for 30 minutes, cool to room temperature, and dilute to the mark with 0.1N sodium hydroxide. Concomitantly determine the absorbances at 225 mu of this solution and of a standard solution of benzoic acid* in 0.1N sodium hydroxide at a similar concentration on a suitable spectrophotometer using matched 1.00 cm silica cells and 0.1N sodium hyroxide as the reference solvent. Calculate the percent of benzoic acid in the powder.

*Benzoic acid standard may be obtained as directed in the "Reagents" section of the U.S.P.

4.3.2.4 Heavy metals.

Dissolve 1 gram of sample in 25 ml of acetone, add 2 ml of purified water and 10 ml of U.S.P. hydrogen sulfide T.S. Any color produced shall not be darker than that of a control containing 25 ml of acetone, 2 ml of standard lead solution prepared as described in the U.S.P., and 10 ml of U.S.P. hydrogen sulfide T.S.

5. PREPARATION FOR DELIVERY

5.1 Shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-186, and as specified herein.

5.1.1 Immediate containers. Shall comply with the following: Shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way be each other.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

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5.2.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.2.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTE: The National Stock Number shall not be preceded by any prefix.

Strip labeling may be used.

Label adhesion test shall not apply.

If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.3 Packaging and packing.

5.3.1 Unit of issue. One bottle containing 100 tablets, as specified, constitutes one unit of issue.

5.3.2 Unit package. At the option of the contractor, each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.3.3 Procedure code. Procedure code No. 4 of PPP-C-186 applies. except in column 4, delete "12/96" and substitute "12/72."

5.4 Marking.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 and 5.5.8.1 of PPP-C-186.

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 and 5.5.8.1 of PPP-C-186.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.3 and 5.4).

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6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-126-3207	METHENAMINE HIPPURATE TABLETS,1 GRAM, 100s

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.4 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1313

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DATE: 10 March 1975

APPROVED NEW DRUG APPLICATION REQUIRED

The supplier of any item(s) listed below must possess, for such item(s), a New Drug Application which has been approved by the Food and Drug Administration.

National Stock Number	Item Identification
6505-00-126-3207	METHENAMINE HIPPURATE TABLETS, 1 Gram, 100s

SSC-1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-37094

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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